

Generation Interconnection Request System Impact Study Report

For Queue Project Q18-1081

Flatlick 500 kV
December 28, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1081. The Interconnection Customer proposes a 345.0 MW wind facility interconnecting at the Flatlick 500 kV substation in the Ohio area. The request was received on October 17, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 264 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$13,847,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
VALEMOUNT to TRISTATE 345 kV line	GAVIN POWER LLC to SPORN 230 kV Line	130.1	1,168	5.8
PENCE to VALEMOUNT 345 kV	SPORN SUBSTATION to GAVIN POWER LLC 230 kV Line	116.9	1,069	6.1
GAVIN POWER LLC 345/230kV Autotransformer	DILLARD-ORANGE to GAVIN POWER LLC 345 kV L	111.4	608	10.4
BURWELL 500 kV/345 kV Transformer Bank	ALLEGHENY ENERGY UNITS 3 4 & 5 345/1 kV Transformer	100.6	1,191	6.2

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
VALEMOUNT to TRISTATE 345 kV	Rebuild line with higher-capacity conductor	\$67,062,000	3% (\$2,230,000)
PENCE to VALEMOUNT 345 kV	Rebuild line with higher-capacity conductor	\$39,520,000	4% (\$1,522,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
GAVIN POWER LLC 345/230kV Autotransformer	Replace transformer with higher-rated unit	\$15,237,000	53% (\$8,087,000)
BURWELL 500/345 kV Transformer Bank	Replace transformer with higher-rated unit	\$33,171,000	6% (\$2,008,000)
	Total allocated to project		\$13,847,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.